Section 1.2: Applications of Linear and Rational Equations

Linear and rational equations are used to model many real-world relationships between quantities. These types of equations allow us to represent costs, motion, and rates of work using algebraic relationships that can be solved systematically.

> **Definition**
>
> A **linear equation** in one variable can be written in the form
>
> $$ax + b = 0,$$
>
> where a and b are constants and $a \neq 0$.

A linear equation represents a direct relationship between two quantities. Solving for x allows us to find the value that satisfies this balance.

Example: Solve $5x - 8 = 12$.

$$5x = 20 \quad \Rightarrow \quad x = 4$$

Example: Solve $7(x - 3) = 14$.

$$7x - 21 = 14 \quad \Rightarrow \quad 7x = 35 \quad \Rightarrow \quad x = 5$$

Real-World Example: Repair Service Charge

A repair technician charges a fixed service fee of \$40 plus \$60 per hour of work. Write and solve a linear model for the total cost if the technician works for 2.5 hours.

Solution:

Let C represent total cost and h represent hours worked. Then $C = 60h + 40$.

$$C = 60(2.5) + 40 = 150 + 40 = 190$$

The total repair cost is \$190.

Example: Solve $3(2x - 5) + 4 = 16$.

$$6x - 15 + 4 = 16 \quad \Rightarrow \quad 6x - 11 = 16 \quad \Rightarrow \quad 6x = 27 \quad \Rightarrow \quad x = \frac{9}{2}$$

Example: Find x if $4(x - 1) = 3x + 2$.

$$4x - 4 = 3x + 2 \quad \Rightarrow \quad x = 6$$

Real-World Example: Cell Phone Plan

A cell phone company charges a monthly fee of \$30 plus \$0.10 per text message. If a customer's monthly bill was \$55, find how many text messages were sent.

Solution:

Let n be the number of messages. Then $0.10n + 30 = 55$.

$$0.10n = 25 \quad \Rightarrow \quad n = 250$$

The customer sent 250 text messages that month.

Definition

A **rational equation** is an equation that contains one or more rational expressions— fractions with polynomials in the numerator and denominator.

To solve a rational equation, first multiply both sides by the least common denominator (LCD) to eliminate fractions, then simplify and solve the resulting linear or quadratic equation. Always check for extraneous solutions caused by restrictions in the denominators.

Example: Solve $\frac{x}{3} + \frac{x}{6} = 5$.

The LCD is 6. Multiply both sides by 6:

$$2x + x = 30 \quad \Rightarrow \quad 3x = 30 \quad \Rightarrow \quad x = 10$$

Example: Solve $\frac{2}{x} + 3 = 5$.

$$\frac{2}{x} = 2 \quad \Rightarrow \quad 2 = 2x \quad \Rightarrow \quad x = 1$$

Always check that x does not make any denominator zero — here $x = 1$ is valid.

Real-World Example: Work Rate Problem

Two printers working together can complete a job in 3 hours. If printer A alone can do the job in 5 hours, how long would printer B take working alone?

Solution:

Let x = time (in hours) for printer B to finish alone. Then printer A's rate is $\frac{1}{5}$ job/hour, and printer B's rate is $\frac{1}{x}$ job/hour.

Together:

$$\frac{1}{5} + \frac{1}{x} = \frac{1}{3}$$

Multiply by $15x$:

$$3x + 15 = 5x$$

$$15 = 2x \quad \Rightarrow \quad x = 7.5$$

Printer B would take 7.5 hours alone.

Example: Solve $\frac{2x}{x-3} = 4$.

Multiply both sides by $x - 3$:

$$2x = 4(x-3) \quad \Rightarrow \quad 2x = 4x - 12 \quad \Rightarrow \quad -2x = -12 \quad \Rightarrow \quad x = 6$$

Check: $x - 3 \neq 0 \Rightarrow x \neq 3$, so $x = 6$ is valid.

Real-World Example: Distance

A car travels 60 miles at 40 mph and returns at 60 mph. Find the total travel time.

Solution:

Use $t = \frac{d}{r}$. Outbound: $t_1 = \frac{60}{40} = 1.5$ hours. Return: $t_2 = \frac{60}{60} = 1$ hour.

$$t_{total} = t_1 + t_2 = 2.5 \text{ hours}$$

Example: Solve $\dfrac{x+2}{x} = 3$.

$$x + 2 = 3x \quad \Rightarrow \quad 2 = 2x \quad \Rightarrow \quad x = 1$$

Extra Practice

1) The cost of renting a car is \$50 plus \$0.20 per mile. Write and solve for 300 miles.

2) Solve for x: $2(x+5) = 3x - 10$.

3) Solve for x: $\dfrac{x}{4} + \dfrac{x}{6} = 5$.

4) A pool can be filled by one pump in 4 hours and drained by another in 6 hours. How long will it take to fill the pool if both pumps are operating?

5) Solve for x: $\dfrac{3}{x+2} = 2$.

6) A car travels 120 miles at 60 mph and returns at 40 mph. Find the total travel time.